CROSS - CORRELATION

THE CROSS-CORRELATION IS A SIMILARITY MEASURE THAT CAN BE USED TO DEFINE TIME DELAYS, TO CHECK HOW SIMILAR TWO SIGNALS ARE TO EACH OTHER AND TO EVALUATE HOW SYNCHRONOUS A SYSTEM IS WITH RESPECT TO A SIGNAL.

THE CONVOLUTION IS: $x[n] *_n y[n] = \sum_{k=-\infty}^{+\infty} x[k]\, y[n-k]$

THE CROSS-CORRELATION IS SLIGHTLY DIFFERENT:

FLIPPED

$$C_{xy}[n] = \sum_{k=-\infty}^{+\infty} x[k]\, y[k-n] = x[n] *_n y[-n]$$

DESCRIBES SIGNALS AND ORDER!

$$C_{xy}[n] = C_{yx}[-n]$$

ANOTHER IMPORTANT PROPERTY IS THAT

$$\left|C_{xy}[n]\right| \leq \left|\sqrt{E_x E_y}\right| \quad \forall n \iff \left|\frac{C_{xy}[n]}{\sqrt{E_x E_y}}\right| \leq 1$$

EX

$x[n] = 2\delta[n-1] + 2\delta[n] + 2\delta[n+1]$

$y[n] = \delta[n-1] + \delta[n] + 2\delta[n+1]$

$$C_{xy}[n] = \sum_{k=-\infty}^{+\infty} x[k]\, y[k-n]$$

$$C_{xy}[0] = \sum_{k=-\infty}^{+\infty} x[k]\, y[k] = 2 + 2 + 4 = 8$$

$x[n] = \{2 \;\; 2 \;\; 2\}$

$y[n] = \{2 \;\; 1 \;\; 1\}$

$$C_{xy}[1] = \sum_{k=-\infty}^{+\infty} x[k]\, y[k-1] = 0 + 4 + 2 + 0 = 6$$

$x[n] = \{2 \;\; 2 \;\; 2 \;\; 0\}$

$y[n] = \{0 \;\; 2 \;\; 1 \;\; 1\}$

$$C_{xy}[2] = \sum_{k=-\infty}^{+\infty} x[k]\, y[k-2] = 4$$

$x[n] = \{2 \;\; 2 \;\; 2 \;\; 0 \;\; 0\}$

$y[n] = \{0 \;\; 0 \;\; 2 \;\; 1 \;\; 1\}$

$C_{xy}[-1] = 4$

$C_{xy}[-2] = 2$

$$C_{xy}[n] = 2\delta[n+2] + 4\delta[n+1] + 8\delta[n] + 6\delta[n-1] + 4\delta[n-2]$$

$$C_{yx}[0] = \sum_{k=-\infty}^{+\infty} y[k]\, x[k] = 4+2+2 = 8$$

$$\begin{array}{l} y[n] = \{2 \;\; 1 \;\; 1\} \\ x[n] = \{2 \;\; 2 \;\; 2\} \end{array}$$

$$C_{yx}[1] = \sum_{k=-\infty}^{+\infty} y[k]\, x[k-1] = 2+2 = 4$$

$$\begin{array}{l} y[n] = \{2 \;\; 1 \;\; 1\} \\ \quad x[n] = \{2 \;\; 2 \;\; 2\} \end{array}$$

$$C_{yx}[1] = \sum_{k=-\infty}^{+\infty} y[k]\, x[k-1] = 2$$

$$\begin{array}{l} y[n] = \{2 \;\; 1 \;\; 1\} \\ \qquad x[n] = \{2 \;\; 2 \;\; 2\} \end{array}$$

$$C_{yx}[-1] = \sum_{k=-\infty}^{+\infty} y[k]\, x[k+1] = 6$$

$$\begin{array}{l} \quad y[n] = \{2 \;\; 1 \;\; 1\} \\ x[n] = \{2 \;\; 2 \;\; 2\} \end{array}$$

$$C_{yx}[-2] = \sum_{k=-\infty}^{+\infty} y[k]\, x[k+2] = 4$$

$$\begin{array}{l} \qquad y[n] = \{2 \;\; 1 \;\; 1\} \\ x[n] = \{2 \;\; 2 \;\; 2\} \end{array}$$

$$C_{yx}[n] = 4\delta[n+2] + 6\delta[n+1] + 8\delta[n] + 4\delta[n-1] + 2\delta[n-2] = C_{xy}[-n]$$

AUTO-CORRELATION

IT IS THE CROSS-CORRELATION OF A SIGNAL WITH ITSELF

$$C_{xx}[n] = \sum_{k=-\infty}^{+\infty} x[k]\, x[k-n]$$

IN ZERO THE AUTO-CORRELATION REPRESENTS THE ENERGY OF THE SIGNAL

$$C_{xx}[0] = \sum_{k=-\infty}^{+\infty} x[k]\, x[k] = \sum_{k=-\infty}^{+\infty} |x[k]|^2 = E_x$$

ANOTHER IMPORTANT PROPERTY IS THAT

$$|C_{xx}[n]| \leq |C_{xx}[0]| \quad \forall n \iff \left| \frac{C_{xx}[n]}{C_{xx}[0]} \right| \leq 1$$

THUS TO CHECK SYNCHRONIZATION IT IS POSSIBLE TO FIND THE MAXIMUM OF C_{xx}